Bryce Grant

Curriculum Vitae

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Research interests: robot manipulation and robot foundation models; mechanistic interpretability of vision-language-action models, world-action models and transformers; training-free planning with runtime verification; 3D vision.

Education

Case Western Reserve University — Cleveland, OH
Ph.D. Candidate, Electrical Engineering; advisor Peng (Edward) Wang — Aug. 2024 – expected May 2028
Advanced to candidacy.

University of Kentucky — Lexington, KY
B.S. Computer Engineering and B.S. Electrical Engineering; minors in Computer Science and Mathematics — May 2024
Magna Cum Laude; Lexmark Scholar.

Honors and Awards

- NSF Graduate Research Fellowship, 2025
- NVIDIA Academic Grant (SPARK), 2025
- Oral presentation, IROS 2025
- Oral presentation, ICLR 2026 Multimodal Intelligence Workshop
- Magna Cum Laude and Lexmark Scholar, University of Kentucky, 2024

Publications (* equal contribution)

Conference papers

- **Not All Features Are Created Equal: A Mechanistic Study of Vision-Language-Action Models**. **B. Grant***, X. Zhao*, P. Wang. *NeurIPS 2026; ICLR 2026 Multimodal Intelligence Workshop (Oral)*.
- **Gluing Local Contexts into Global Meaning: A Sheaf-Theoretic Decomposition of Transformer Representations**. **B. Grant**, P. Wang. *NeurIPS 2026*.
- **Quaternion Approximate Networks for Enhanced Image Classification and Oriented Object Detection**. **B. Grant**, P. Wang. *IROS 2025 (Oral)*.

Preprints

- **SPARK: Sequential Planning via Anchored Robotic Keypoints**. **B. Grant**, A. Rothenberg*, L. Senning*, P. Wang. *Under review, ICLR 2027; arXiv:2606.30613*.
- **TrianguLang: Geometry-Aware Semantic Consensus for Pose-Free 3D Localization**. **B. Grant**, A. Rothenberg, A. Banerjee, P. Wang. *Under review; arXiv:2603.08096*.

Research Experience

Case Western Reserve University, AISM Lab — Cleveland, OH
Graduate Research Assistant (advisor Peng (Edward) Wang) — Aug. 2024 – Present

- **SPARK (lead):** training-free manipulation; an LLM-planned behavior tree anchored to live 3D keypoints, verified per primitive with local recovery. 75.1% on LIBERO-PRO, 66.7% with objects moved 10 cm mid-task, 68% over 11 physical settings on UR10e, Franka FR3 and a bimanual rig.
- **VLA interpretability (co-lead):** 520+ sparse autoencoders over $\pi_{0.5}$, OpenVLA-OFT, X-VLA, SmolVLA and GR00T N1.5 ($\sim$10K A100-hours); activation injection shows policies bind motor programs to workspace coordinates (99.6% of $\pi_{0.5}$ episodes; 26/26 on a real UR10e). Released 7 TB of data and Action Atlas.
- **World-action models:** swapping only the imagined frames' keys and values redirects LingBot-VA (39/40) and DreamZero (40/40) as a new instruction would; the target is read at the last denoising steps, and masking the imagined future lowers success on all eight WAMs tested.
- **Horizon tomography:** under a kernel control of the present, six policies' apparent knowledge of the future reduces to the robot's own upcoming motion, within a 0.02–0.03 equivalence bound set by planted signals.
- **Sheaf interpretability:** a training-free cellular sheaf on paraphrase graphs; its 20-dim kernel beats LEACE by +17.1 pp on held-out fact retrieval (8 LLMs), and its complement localizes causal influence (9 models, 124M–13B).
- **TrianguLang:** calibration-free, feed-forward 3D localization from a text query with geometry-gated cross-view attention; state of the art among feed-forward methods on seven benchmarks at $\sim$17 FPS.
- **SPARK-Tempo:** timed-execution testbed in MuJoCo-Warp with Isaac Sim cross-checks and audio/tactile verification (guitar playing on a Unitree G1); custom Warp kernels.

- Research with AI coding agents under hard validation: agent-written audits caught a leak in my own benchmark harness (disclosed and rerun); a 1,300-test CI gates the SPARK codebase.
- Mentored three undergraduate researchers, all co-authors on SPARK or TrianguLang.
- **QUAN:** quaternion convolution layers for rotation-aware image classification and oriented object detection with far fewer parameters (IROS 2025 oral).

University of Kentucky, AI for Smart Manufacturing Lab Lexington, KY
Undergraduate Researcher Aug. 2023 – May 2024

- CNN-Transformer hybrid for 6-DoF pose estimation from RGB-D for robot-arm manipulation.
- Led navigation for an autonomous telepresence robot: 3D SLAM with a servo-mounted LiDAR and dynamic obstacle avoidance.

Industry Experience

Mercor Remote
Applied AI Engineer Intern Aug. 2025 – Aug. 2026

- Built RL environments, synthetic-data pipelines and trajectory-annotation tools for coding-agent post-training (RLHF/SFT) programs; set acceptance criteria for agent runs.
- Built a QA pipeline for egocentric manipulation video datasets and a failure-mode taxonomy of frontier-model evaluation trajectories that shaped post-training data strategy.
- PySpark/Snowflake pipelines tracking throughput, task accuracy and budget across projects.

HP Vancouver, WA
PhD ML Intern May – Aug. 2024

- Benchmarked retrieval-augmented generation: semantic vs. recursive chunking, BM25 vs. FAISS retrieval and LLM query expansion, evaluated with coherence and NDCG.
- Built a customer-facing LLM agent for printer setup with dynamic reranking and multi-stage error recovery.
- Deployed an anomaly-detection pipeline on AWS for enterprise cloud spend (LSTM forecasting with STL decomposition).

Honeywell Atlanta, GA
Embedded Software Engineer Intern May – Aug. 2023

- Bare-metal firmware API for life-safety microcontrollers on a product line of 100K+ units.
- Validated dual-CPU communication protocols with oscilloscope debugging and hardware-in-the-loop testing.

Talks

- *From Seeing to Acting to Imagining: The Evolution of Robot Foundation Models*. Invited talk, Free University of Bozen-Bolzano, Oct. 2026.
- *Not All Features Are Created Equal*. Oral, ICLR 2026 Multimodal Intelligence Workshop, Rio de Janeiro, Apr. 2026.
- *Quaternion Approximate Networks*. Oral, IROS 2025, Hangzhou, Oct. 2025.

Mentoring, Service and Leadership

- Mentored three undergraduate researchers (A. Rothenberg, L. Senning, A. Banerjee), all co-authors on SPARK or TrianguLang.
- Reviewer: CoRL 2026, ICLR 2026–2027, ECCV 2026, IROS 2025–2026, NAMRC 2025–2026.
- MathWorks Student Ambassador, Oct. 2024 – Dec. 2025: hosted hackathons and educational events.
- NSBE Region III Finance Chair, May 2023 – May 2024: organized a regional career fair with 50+ companies and raised over $250,000.

Skills

- **Programming:** Python, C/C++, CUDA, Warp, MATLAB, SQL
- **ML:** PyTorch, JAX, HuggingFace/LeRobot, vLLM, distributed training, sparse autoencoders, flow-matching policies, VLAs ($\pi_{0.5}$, OpenVLA-OFT, GR00T, SmolVLA, X-VLA), world-action models (Cosmos Policy, DreamZero), RL (PPO), Weights & Biases
- **Robotics and simulation:** MuJoCo, MuJoCo-Warp, Isaac Sim/Lab, mjlab, robosuite/LIBERO, RoboTwin 2.0, ROS 2, ur_rtde, franky, cuRobo, OpenCV, Open3D, COLMAP, SAM3, Depth Anything 3
- **Platforms:** UR10e/UR5e with Robotiq 2F-85, Franka FR3/Panda (incl. bimanual), Unitree Go2 with D1 arm and G1, ORCA hand (sim), FlexiTac tactile, Azure Kinect, RealSense, ZED
- **Infrastructure:** Docker, Linux, Slurm (NCSA Delta, Purdue Anvil), Modal, PySpark, Snowflake, Blender